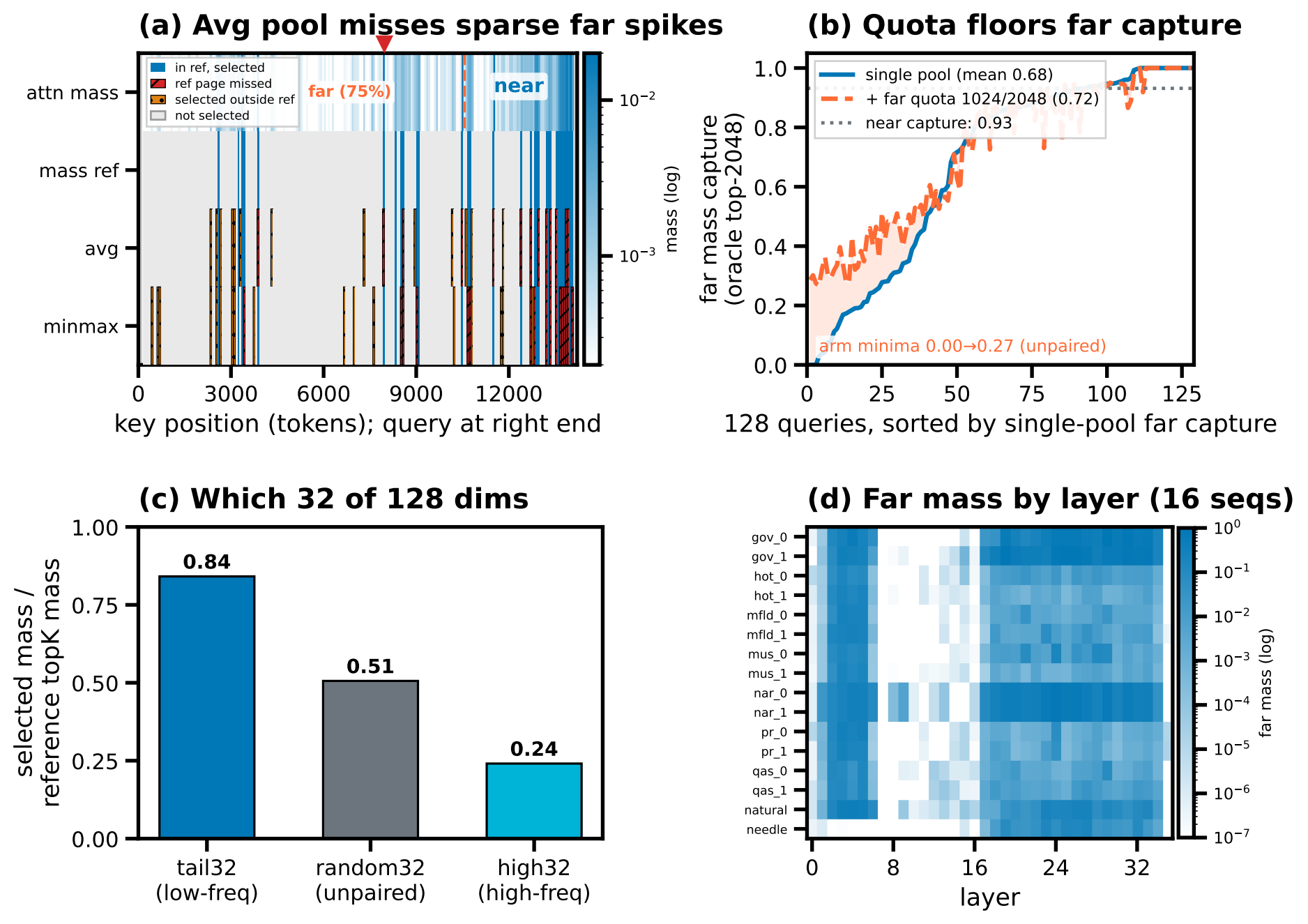

(a) Avg pool misses sparse far spikes
in ref, selected
ref page missed
selected outside ref
not selected
far (75%)
near
attn mass
mass ref
avg
minmax
10^{-2}
10^{-3}
mass (log)
0
3000
6000
9000
12000
key position (tokens); query at right end
(b) Quota floors far capture
single pool (mean 0.68)
+ far quota 1024/2048 (0.72)
near capture: 0.93
arm minima 0.00→0.27 (unpaired)
far mass capture
(oracle top-2048)
1.0
0.8
0.6
0.4
0.2
0.0
0
25
50
75
100
125
128 queries, sorted by single-pool far capture
(c) Which 32 of 128 dims
selected mass /
reference topK mass
1.00
0.75
0.50
0.25
0.00
0.84
0.51
0.24
tail32
(low-freq)
random32
(unpaired)
high32
(high-freq)
(d) Far mass by layer (16 seqs)
gov_0
gov_1
hot_0
hot_1
mfld_0
mfld_1
mus_0
mus_1
nar_0
nar_1
pr_0
pr_1
qas_0
qas_1
natural
needle
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
far mass (log)
0
8
16
24
32
layer